Sahaj Bhadja

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Summary

Software product test engineer with 2+ years authoring **automated test scripts**, executing formal product testing, and tracking defects across embedded-style systems. Proficient in **Python**, **C/C++**, **MS Office**, QA practices, and **AI-assisted** test development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Software Engineering, Operating Systems, Database Systems, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, OOP, Operating Systems, Data Structures

Experience

USC Dornsife **Los Angeles, CA**
Software Product Test Engineer *September 2025 – Present*
- Analyzed functional requirements to generate **test cases** for multi-tenant workflows serving **500+ users**, executing validation checks, reporting defects, and verifying fixes across Azure SQL integration pipelines in production environments.
- Wrote automated validation scripts and performance **test procedures** for FastAPI jobs processing **5K+ monthly workloads**, advancing QA processes that achieved **99%+ reliability** while meeting short- and long-term delivery deadlines.

Grad8 **Los Angeles, CA**
Software Product Test Engineer Intern *June 2025 – August 2025*
- Developed functional and regression **test procedures** for AI-assisted workflows handling **10K+ daily requests**, conducting exploratory testing, reporting issues, and using **generative AI tools** to accelerate test script development.
- Executed integration testing on AWS RDS modules with GitHub Actions automation, verifying defect fixes and communicating results to internal customers in a collaborative team environment under tight release schedules.

Aasma Technology Solutions **India**
Software Product Test Engineer - (SecureSync) *August 2023 – July 2024*
- Authored functional **test cases** and **Python automated scripts** across **20+ PyTest suites**, executing formal testing on VM **test facilities** spanning Windows, Linux, and macOS while reporting defects through Jenkins tracking workflows.
- Maintained engineering **test infrastructure** including nightly automation and VM **simulators**, verifying fixes and recommending process improvements that increased release confidence by **68%** in Agile Scrum teams.

Aasma Technology Solutions **India**
Software Product Test Engineer Intern - (VibrantChem) *May 2023 – August 2023*
- Executed integration and exploratory testing on real-time modules syncing **2M+ records**, writing **Jest** and **Selenium automated scripts** while supporting Docker/Jenkins pipelines and meeting customer reliability needs in team environments.
- Conducted component-based validation across authentication, billing, and resource workflows, tracking defect resolution and assisting test software maintenance while adhering to safety-conscious distributed system testing practices.

MedDevice Test Systems **India**
Software Product Test Engineer Intern *January 2023 – April 2023*
- Contributed to **embedded controller** testing with deterministic **simulators** and **Python/C++** libraries, configuring device scenarios, logging defects, and verifying fixes across hardware-software integrated control workflows.
- Integrated test coverage reports into CI pipelines, helping maintain test hardware/software setups and advancing automated testing tools while communicating defect status with engineering partners verbally and in writing.

Skills

Testing: test case development, functional/regression/integration/exploratory testing, automated testing, test procedures, defect tracking, QA, test methodologies, simulators, test facilities, component-based development
Technical: Python, C, C++, PyTest, Jenkins, Jest, Selenium, Git, Linux, Windows, OOP, embedded controls (familiar), building/industrial controls (familiar)
Tools: MS Office, Excel macros, pivot tables, Excel formulas, XML, MS Project (familiar), generative AI, AI-assisted development, Agile, customer communication

Projects

CPU-scheduling Simulator Live | *C++, Simulators, Automated Validation*
- Built a **C++** scheduling **simulator** implementing FCFS, SJF, RR, and MLQ algorithms, executing automated correctness tests across **1K+ process combinations** and reducing debugging time by **20%**.

Big-Brain Live | *Python, AI Tools, Integration Testing*
- Developed an AI-assisted platform using **generative AI**, writing validation scripts and integration tests across ingestion pipelines processing **50K+ records** with structured defect reporting workflows.

ControlsSim | *C++, Python, Embedded-Style Controller Testing*
- Created a control-loop **simulator** emulating sensor/actuator states, authoring **Python** and **C++** automated scripts validating state transitions, fault handling, and controller configuration scenarios.